Tremorlog

EXECUTIVE SUMMARY

Tremorlog, a student startup from our university

The Problem

Parkinson's disease is a progressive neurological disorder which effects close to one million people in the United States, and tremor is one of the most visible and disruptive symptoms. The problem is that the way doctors currently track it is very outdated. A patient with Parkinson's typically sees their neurologist every three to six months, and the appointment is short. During that time the doctor observes the patient, asks questions, and makes a judgement about how medication is working. But Parkinson's symptoms fluctuate alot during the day and from week to week, depending on medication timing, sleep, stress and many other factors. So what the neurologist sees in the room is only a snapshot, and often not a representative one. Patients are asked to remember how they did over the past months, and memory is unreliable especially when the disease can also effect cognition. Some patients keep paper diaries but these are inconsistent and subjective. As a result medication decisions are sometimes made with incomplete information, and worsening symptoms may go unnoticed for months. There is a real gap in remote monitoring of movement disorders, unlike conditions such as diabetes where home devices provide continuous data.

Our Solution

Tremorlog is a web application that fills this gap in a simple way. The patient, or a caregiver, opens the app on a phone, tablet or laptop and records a short video, around thirty seconds, of their hands at rest and holding a posture, following simple on screen prompts. This takes about two minutes and is meant to be done once a day at a consistent time. Our software then processes the video using computer vision to estimate tremor amplitude and frequency and generates a daily score. Over time the app builds a trend line which the patient can see too. On the other side, the neurologist has a clinician dashboard that shows each patient's trend, flags significant changes, and lets them review the actual video clips if they want to. Before an appointment the doctor can see weeks of data rather than relying on a single observation. We think this makes visits more productive and could allow earlier adjustments to treatment. We chose a web app rather than a native mobile app so that there is nothing to download, which matters because many Parkinson's patients are older and not always comfortable with technology.

Market

There are approximately one million people living with Parkinson's in the US and about sixty thousand new diagnoses each year. Beyond Parkinson's the same approach could extend to essential tremor and other movement disorders which would make the market several times bigger. There are also roughly several thousand practicing neurologists and movement disorder specialists. The remote patient monitoring market is growing quickly, driven by changes in how telehealth is reimbursed and by greater acceptance from both patients and clinicians after the pandemic. We have not yet done a full bottom up market sizing but if even a small percentage of Parkinson's patients were paying a monthly fee through their clinic this would be a meaningful business. Our initial target is small and mid sized neurology practices and movement disorder clinics because they can make purchasing decisions faster than large hospital systems.

Business Model

Our plan is to sell subscriptions to clinics, not to individual patients. The clinic pays a monthly fee per active patient enrolled, which we are currently testing at a price in the range of fifteen to twenty five dollars per patient per month. This is partly justified because remote monitoring services may be billable by clinics under existing reimbursement codes, although we have not yet confirmed exactly which codes apply to our product and this is something we need to research with a healthcare reimbursement expert. Secondary revenue could come later from anonymized data partnerships with pharmaceutical companies running clinical

trials who need objective symptom measures, but this raises privacy questions which we will address carefully and only with patient consent. Our costs are mainly cloud hosting and video storage, which are significant because video files are large, plus development time.

Competition

There are other players in this space. Some companies have wearable sensors that measure tremor, and one or two larger companies have FDA cleared devices worn on the wrist. These are more clinically validated than we are, but they require buying and shipping hardware, are expensive and patients must wear them. There are also general telehealth platforms which allow video visits but do not give continuous or quantitative data. Finally there are symptom diary apps which are cheap but rely on self reporting. Our advantage is that we need no special hardware, only a camera that the patient already owns, and so our cost to deploy is very low. Our weakness is that video based measurement is probably less accurate than a dedicated sensor, and we have not yet formally validated our scores against clinical rating scales. We know this is the biggest question mark about our product.

Team

Priyanka Venkataraman leads product and clinical relationships. She is a senior studying neuroscience and has volunteered at a movement disorders clinic, where the idea started after she saw how little information doctors had between visits. Desmond Okafor-Reyes is our lead engineer, studying computer science, and he built the prototype including the video analysis pipeline. Callum Tishkoff handles business operations, finance and the company registration, and is studying business administration. We do not currently have a physician on the team, but we have been in conversation with a neurologist who has agreed to informally advise us. We recognize that a clinical advisor and someone with regulatory experience are needed.

Milestones and Financials

So far we have defined the core concept, built a working prototype that handles video capture, scoring and the clinician dashboard, and registered the company. Our first client, a small neurology practice, has paid for a pilot covering a limited number of patients, which gave us our first revenue and a lot of useful feedback. The legal entity itself is still not fully formed, and we need to sort out the formal structure and ownership split between founders. Revenue to date is very small, only the pilot fee, and costs have been mostly covered by the founders personally plus small savings. We project that with ten clinics and around two hundred patients we could reach approximately fifty thousand dollars in annual recurring revenue, though this depends on assumptions about pricing and adoption that we have not tested widely. We also do not yet have a website.

What We Are Asking For

We are asking for prize funding to support three things. First, to run a validation study comparing our tremor scores with standard clinical ratings, ideally with the help of a university medical partner. Second, to get legal advice on forming the company properly and on privacy and health data rules including what compliance is needed for handling patient video. Third, to build a basic website and expand the pilot to several more clinics. Beyond money we would value mentorship from people with healthcare startup experience, introductions to neurologists, and guidance on the regulatory pathway. We believe Tremorlog addresses a real and underserved need and we are committed to working on it after graduation.